

Local Violence and Leader Favorability in Mali

Kevin Linares

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1 Introduction

This report estimates whether local political violence is associated with favorable views of the national leader. Violence comes from ACLED event counts aggregated to the commune (admin-3) by wave; favorability comes from three independent survey waves, each drawn under its own stratified two-stage design. The analysis follows the same logic as a sequential multilevel build: variance partitioning, fixed-effect blocks added one at a time, diagnostics, then sensitivity checks that bring the survey design back in.

The questions are:

1. Is a higher count of violent events in a respondent's commune associated with a lower (or higher) probability of a favorable view, net of wave, region, and respondent characteristics?
2. Does that association reflect **change** within a commune over time, or **level** differences between communes?
3. How much does the estimate move when the model accounts for the sample design (strata, PSUs, weights) versus a model-based analysis that does not?
4. Is the association carried through attitudes that violence plausibly shifts (perceived insecurity, stress, local government approval)?

2 Method

2.1 Sample and design

The analysis file contains 5,126 respondents (6.1% removed by listwise deletion from 5,460), nested in 852 wave-specific PSUs and 276 communes across 8 regions. Each wave is a separate probability sample: regions are strata, villages (rural communes) or quarters (urban communes) are PSUs selected with probability proportional to size, and respondents are selected within households. Because PSUs are redrawn each wave, communes enter the data unevenly (Figure 1).

Table 1: Sample and design summary by wave. Each wave is analyzed with its own strata, PSUs, and weights.

Wave	n	PSUs	Com- munes	Design df	% favorable (weighted)	SE	Deff (total)	Deff (Kish)
1	1686	284	209	276	52.9	1.5	1.49	1.36
2	1731	284	210	276	59.3	1.6	1.78	1.36
3	1709	284	202	276	60.6	1.4	1.49	1.38

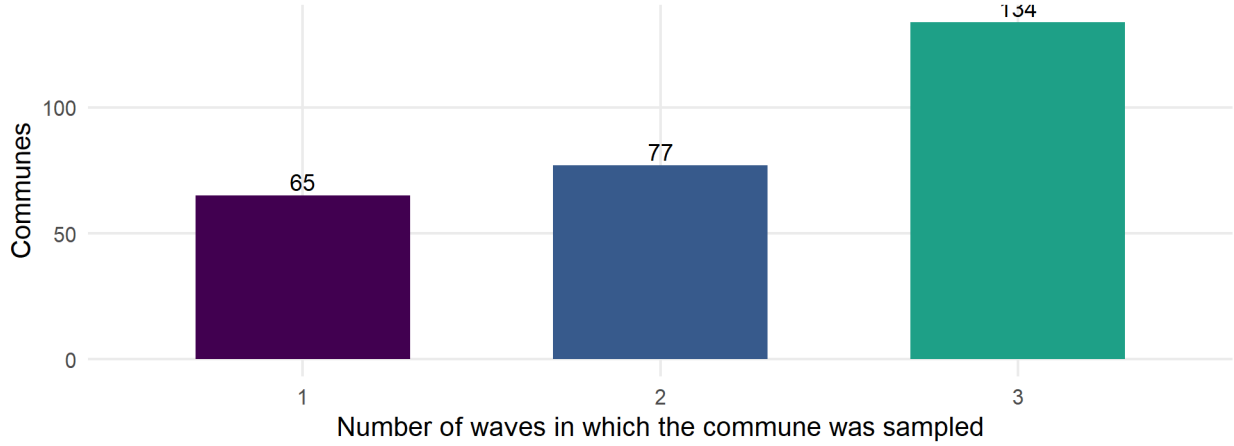


Figure 1: Communes observed in one, two, or three waves. Communes seen more than once contribute within-commune change; all communes contribute between-commune contrasts.

Two implications follow. First, strata and PSU identifiers are made wave-specific (region $\times$ wave, PSU $\times$ wave), so the pooled design is three independent designs stacked, not one design observed three times. Second, weights from different waves sum to different population totals, so for pooled analyses they are rescaled to sum to each wave's sample size, $w_i^* = w_i n_w / \sum_{i \in w} w_i$, which keeps any single wave from dominating.

2.2 Measures

Outcome. $y_i = 1$ if the respondent holds a favorable view of the leader.

Exposure. For commune c in wave w , $V_{cw} = \log(1 + E_{cw})$, where E_{cw} is the ACLED event count in the window before fieldwork. The log compresses a heavily right-skewed count in which most commune-waves are zero. Commune-waves absent from ACLED are set to zero events. For the within/between decomposition, the commune mean uses **all three waves of ACLED data**, whether or not the commune was sampled in that wave:

$$V_c^- = \frac{1}{3} \sum_{w=1}^3 V_{cw}, \quad V_{cw}^{\text{within}} = V_{cw} - V_c^-$$

Covariates. Demographics (female, age in decades, ethnicity, education), urban residence, and two attitudinal controls (local government approval, support for democracy, centered). Perceived violence and stress are treated as **possible mediators** and enter only in a separate model.

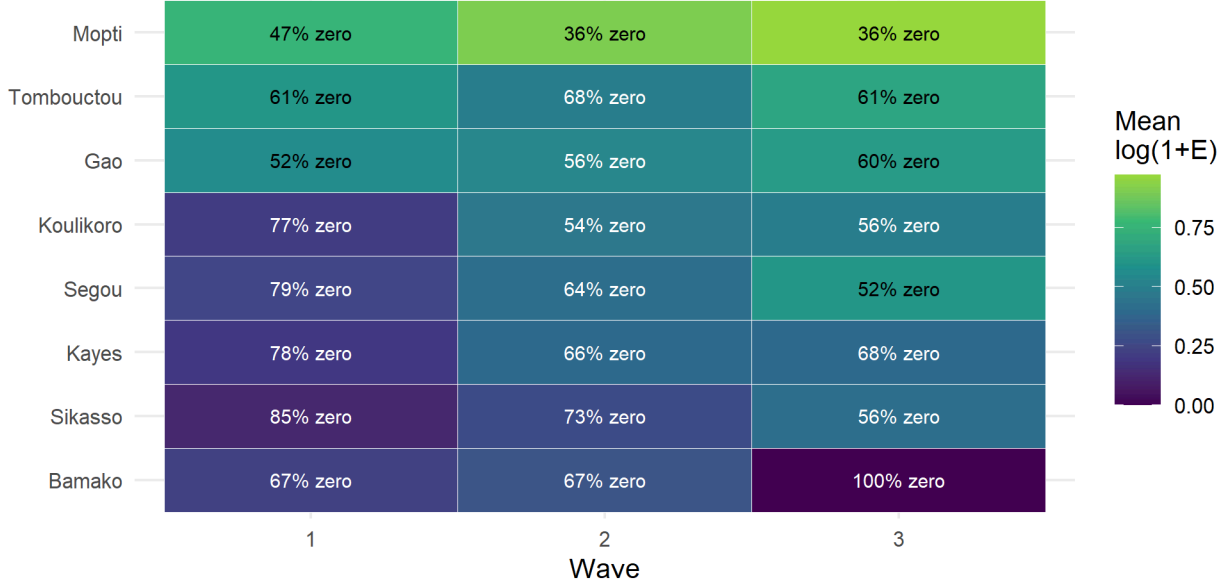


Figure 2: Distribution of commune-level violence by region and wave. Most commune-waves record no events, which is why the exposure is $\log(1 + \text{events})$.

2.3 Model specification

The main model is a two-level random-intercept logistic regression with respondents i in PSU-waves p in communes c :

$$\begin{aligned} \text{logit Pr}(y_{ipcw} = 1) = & \beta_0 + \beta_V V_{cw} + \sum_{w=2}^3 \delta_w \text{Wave}_w + \sum_r \rho_r \text{Region}_r \\ & + \mathbf{x}_i' \boldsymbol{\beta} + u_c + v_p \\ u_c \sim & N(0, \tau_c^2), \quad v_p \sim N(0, \tau_p^2) \end{aligned}$$

Wave and region enter as **fixed effects**. With three waves and eight regions, random effects at those levels would rest on too few units to estimate a variance, and the eight regions are the full set of strata rather than a sample of regions. The commune intercept u_c uses a commune code that is stable across waves, so repeatedly sampled communes share one intercept. The PSU intercept v_p is wave-specific. Intraclass correlations are reported on the latent logistic scale, $\text{ICC}_c = \tau_c^2 / (\tau_c^2 + \tau_p^2 + \pi^2/3)$.

The within/between model replaces $\beta_V V_{cw}$ with $\beta_B V_c^- + \beta_W V_{cw}^{\text{within}}$ (Bell & Jones, 2015). β_W is identified only from communes whose violence changed across waves, which gives it stronger protection against stable commune-level confounding than β_V .

Because logistic coefficients are not on a natural scale and change when covariates are added even without confounding, the headline quantity is the **average marginal effect** (AME) of violence on the probability scale, computed with random effects set to zero:

$$\text{AME}_V = \frac{1}{n} \sum_{i=1}^n \frac{\partial \text{Pr}(y_i = 1 \mid \mathbf{x}_i, u = 0, v = 0)}{\partial V_{cw}}$$

2.4 Model building and estimation

Models were built sequentially on a common analysis file so that likelihood ratio tests (LRTs) are valid: (M0) null model for variance partitioning; (M1) wave and region; (M2) violence; (M3) demographics and urban residence (**main model**); (M4) attitudinal controls; (M5) mediators. M6 is the within/between version of M3. Models were fit by maximum likelihood (Laplace approximation) in `lme4` with the BOBYQA optimizer. AMEs and their delta-method standard errors come from `marginalEffects`.

Design sensitivity. The main models are model-based and unweighted: they condition on the design variables (region as stratum, PSU and commune clustering) rather than weighting. As a check, the same fixed-effects specification is estimated as a design-based logistic regression, `survey::svyglm`, with wave-specific strata and PSUs, rescaled weights, and Taylor-linearized standard errors on 828 design degrees of freedom. Large differences between the two would point to informative sampling or misspecification (DuMouchel & Duncan, 1983). A Bayesian fit in `brms` is available if `lme4` does not converge.

3 Results

3.1 Variance partitioning and model comparison

In the null model, 4% of latent variance in favorability lies between communes and 2% between PSUs within communes. Adding violence to wave and region improved fit, $\chi^2(1) = 40.6$, $p < .001$. Table 2 lists every model.

Table 2: Sequential model comparison. LRTs compare each model with the one before it (M0 to M5); M6 is not nested and is compared by information criteria only. ICCs are on the latent logistic scale.

Model	Purpose	ICC	commune/ PSU	AIC	BIC	LRT vs previous
M0	Null: variance partitioning	0.04	0.02	6924.3	6943.9	
M1	1. wave, region	0.02	0.02	6878.2	6956.7	64.1 (9), $p < .001$
M2	1. violence	0.02	0.01	6839.6	6924.6	40.6 (1), $p < .001$
M3	1. demographics, urban (main)	0.02	0.01	6808.8	6965.8	52.8 (11), $p < .001$
M4	1. local gov, democracy	0.01	0.01	6592.1	6762.2	220.7 (2), $p < .001$
M5	1. perceived violence, stress	0.02	0.01	6570.7	6753.8	25.4 (2), $p < .001$
M6	M3, violence within/between	0.02	0.01	6807.0	6970.5	

3.2 Violence and favorability

In the main model (M3), a one-unit increase in $\log(1 + E)$, roughly the difference between no events and two events, is associated with a 7.6 percentage-point lower probability of a favorable view (95% CI -9.7 to -5.4), an estimate statistically distinguishable from zero ($p < .001$). On the odds scale this is an odds ratio of 0.72 (95% CI 0.66 to 0.79). Moving a commune from zero to three events changes the predicted probability by -10.7 points (95% CI -13.9 to -7.6).

The within/between split (M6) separates the two sources of that association. The **within-commune** AME, identified from communes whose violence changed across waves, is -9.5 points (95% CI -12.5 to -6.6); the **between-commune** AME is -5.3 points (95% CI -8.4 to -2.1). The within estimate is the more defensible

causal quantity because it holds stable commune characteristics fixed; the between estimate is more exposed to confounding by anything that differs across communes.

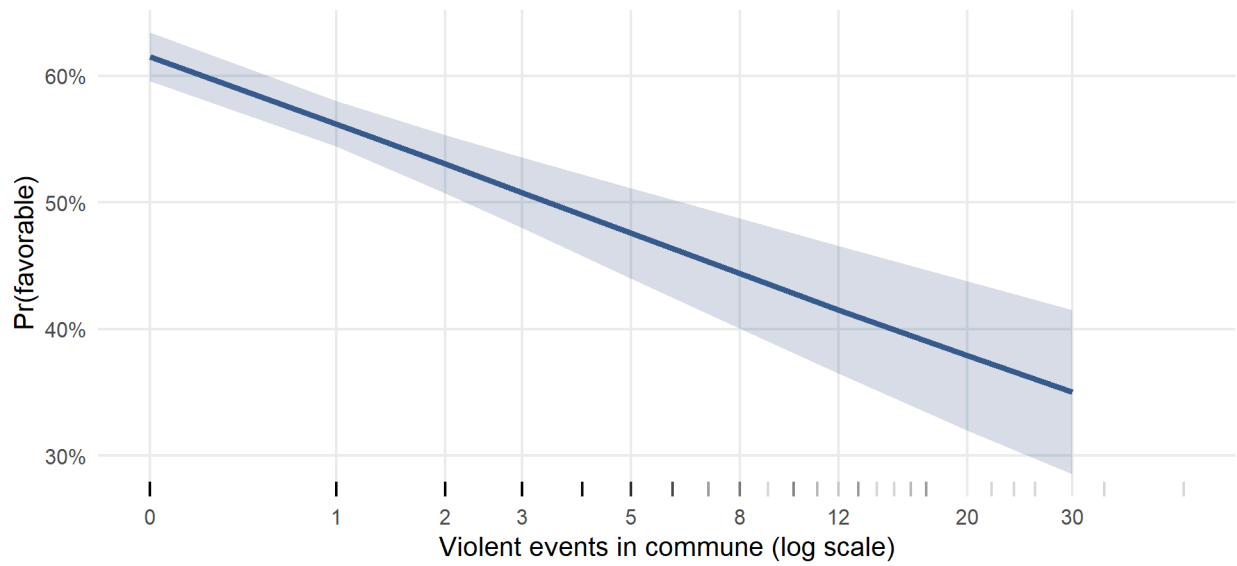


Figure 3: Average predicted probability of a favorable view across commune event counts (main model, random effects at zero), with 95% confidence band. Tick marks show the observed distribution of events.

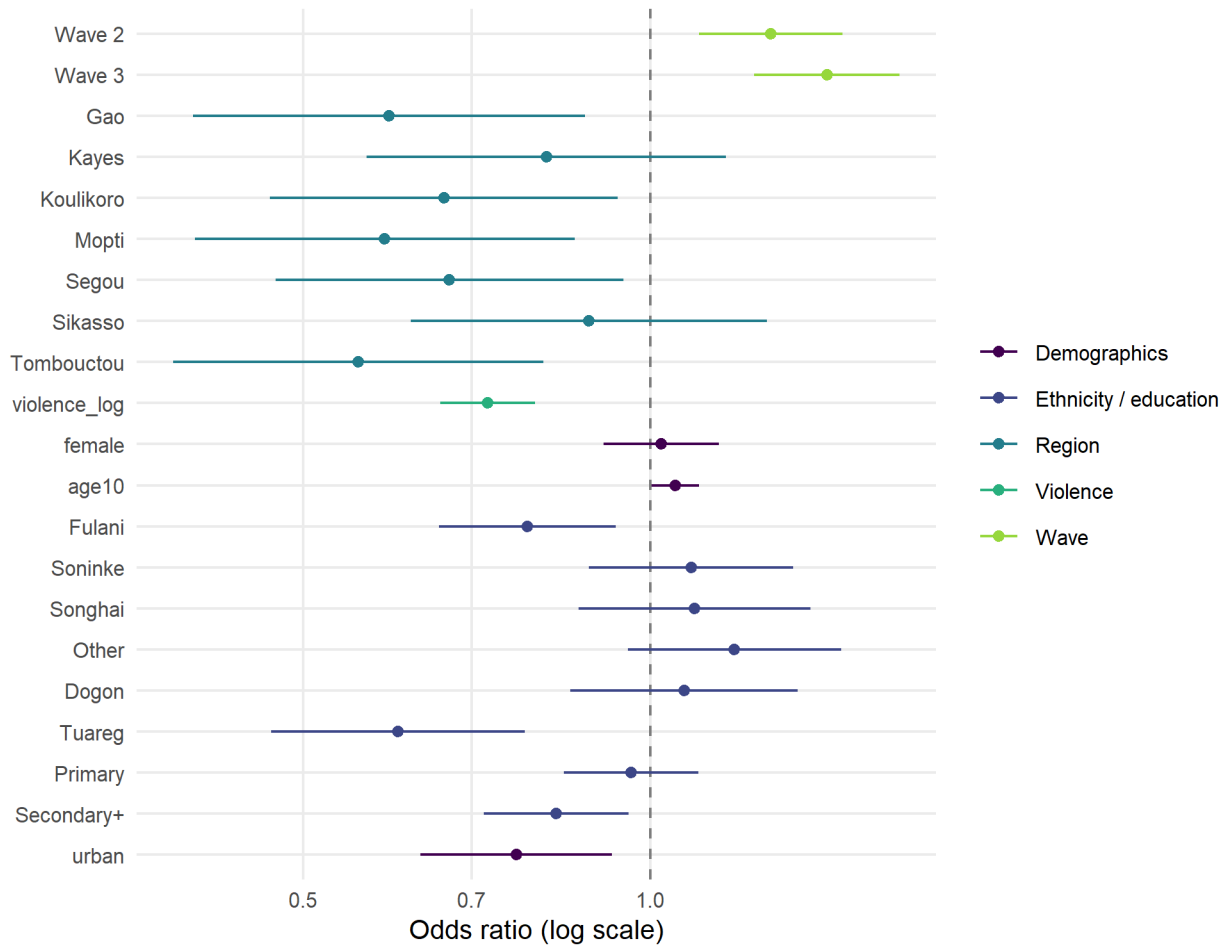


Figure 4: Odds ratios from the main model (M3) with 95% confidence intervals. Reference categories: wave 1, the reference region, the most common ethnic group, and the first education level.

3.3 Pathways through attitudes

Figure 5 shows how the violence AME changes as attitudes enter. From M3 to M5 the estimate changes by 25%. This pattern is consistent with part of the association running through perceived insecurity, stress, and local government approval, but it is **not** a mediation estimate: those attitudes are measured at the same time as the outcome, and any unmeasured cause of both attitudes and favorability biases the comparison. M3 is therefore reported as the total association; M4 and M5 are descriptive.

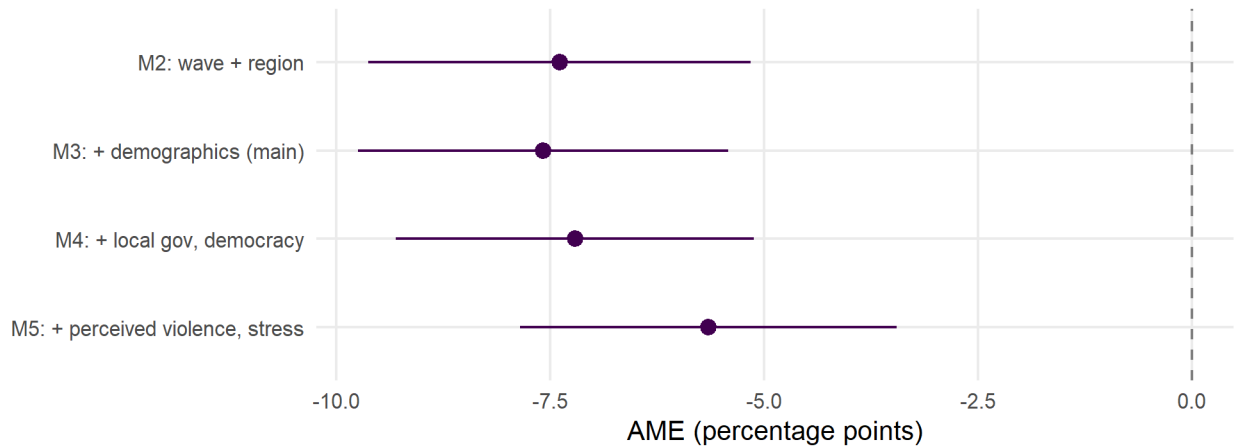


Figure 5: Average marginal effect of violence (percentage points per unit of $\log(1 + \text{events})$) as attitudinal blocks are added.

3.4 Sensitivity: design, exposure, and influential regions

Figure 6 collects the violence AME across specifications. The design-based estimate differs from the model-based estimate by 0.5 points, and its standard error is 1.2 times as large. A difference in point estimates beyond sampling error would suggest the weights carry information the model does not; a larger design-based standard error is expected and reflects clustering and unequal weighting (Table 1). Using a binary exposure, any event versus none, the AME is -9.4 points (95% CI -12.6 to -6.3). Dropping one region at a time, the AME ranges from -8.2 to -7.2 points.

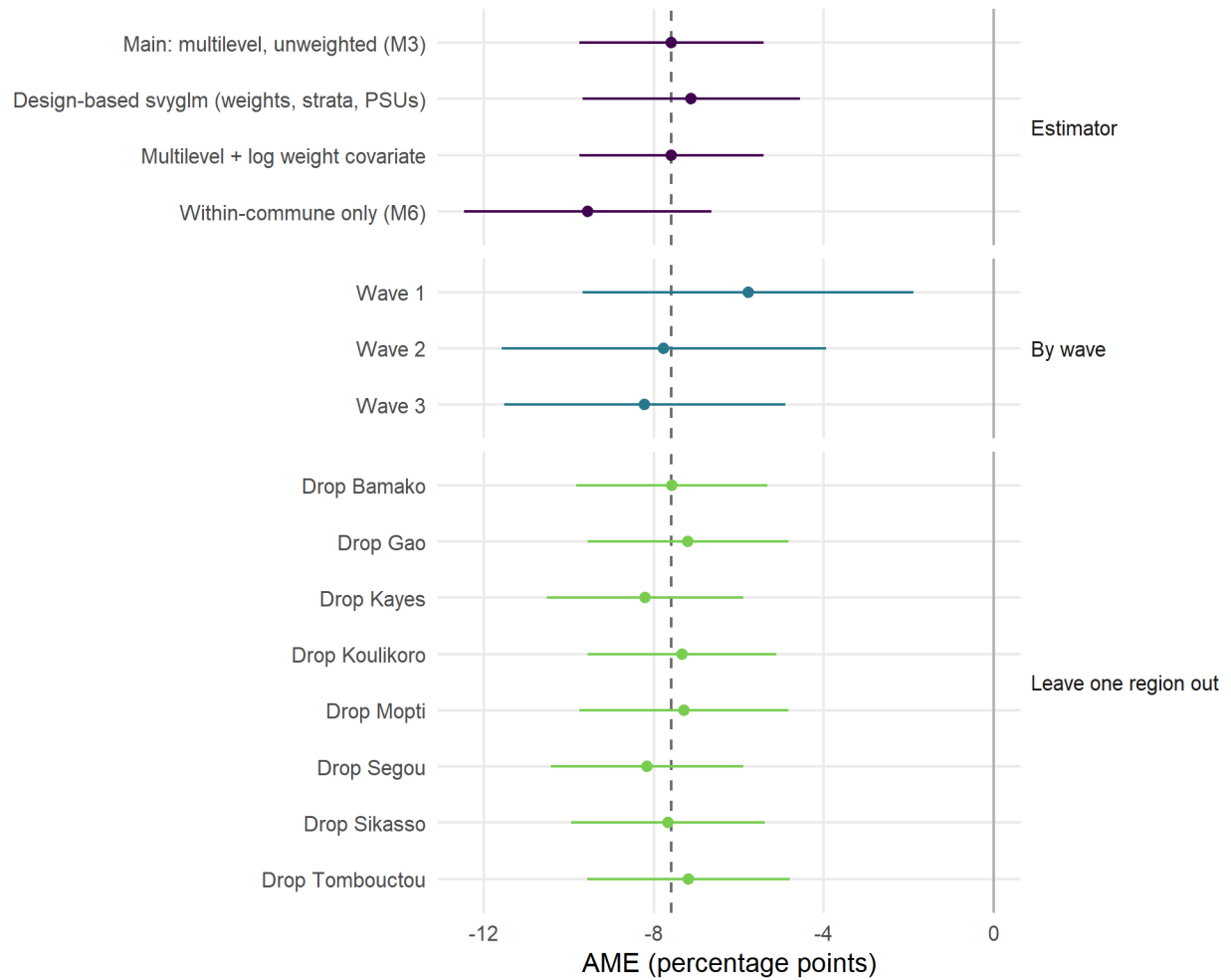


Figure 6: Violence AME (percentage points per unit of $\log(1 + \text{events})$) across estimators, waves, and leave-one-region-out fits. The dashed line marks the main-model estimate.

3.5 Transparency check: region-wave means

Figure 7 strips the model away: each point is a region in a wave, with design-based favorability plotted against the average commune violence of its respondents. If the multilevel estimate is driven by a few extreme cells rather than a consistent pattern, it shows up here.

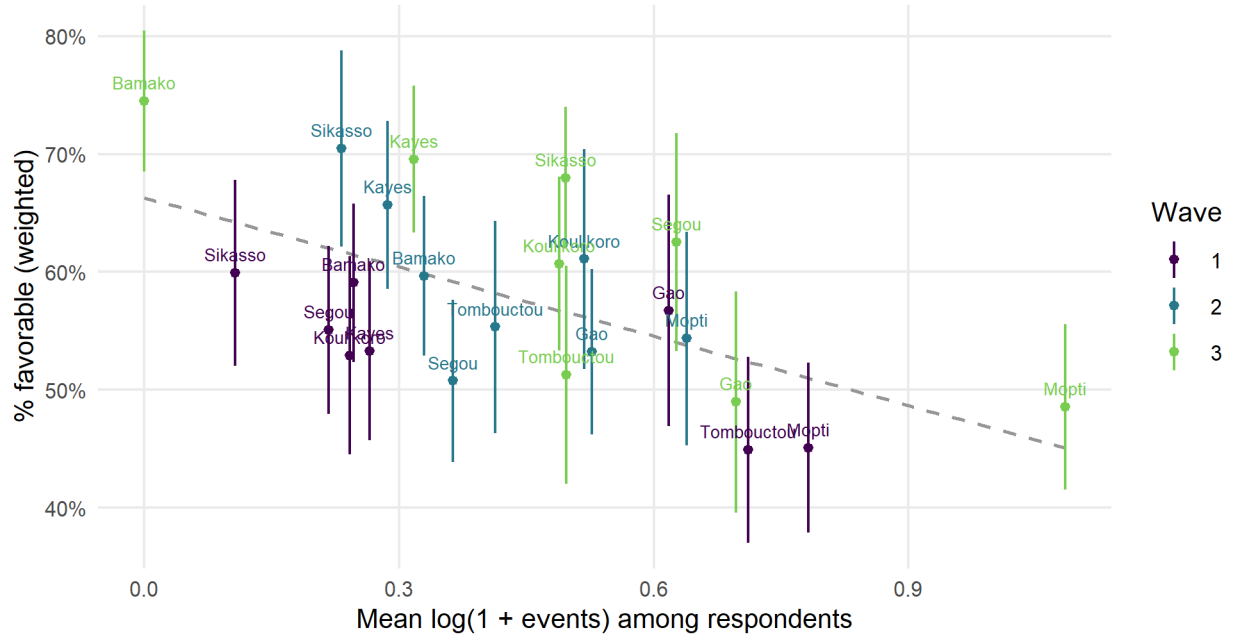


Figure 7: Design-based percent favorable (with 95% CI) against mean respondent exposure, for each region and wave.

3.6 Geography of favorability and violence

Figure 8 puts both quantities at the commune level. Violence is mapped for all 696 communes, because ACLED covers the whole country. Favorability can only be mapped where the survey sampled, 0 communes across the three waves, and unsampled communes are left gray. Favorability is the model-based estimate from the main model rather than the raw commune mean: with a handful of respondents per commune, raw means would mostly display sampling noise, while the model estimate shrinks small communes toward their region-wave expectation. Commune-level design-based estimates are not shown because most communes contain only one or two PSUs, too few for a stable design-based variance.

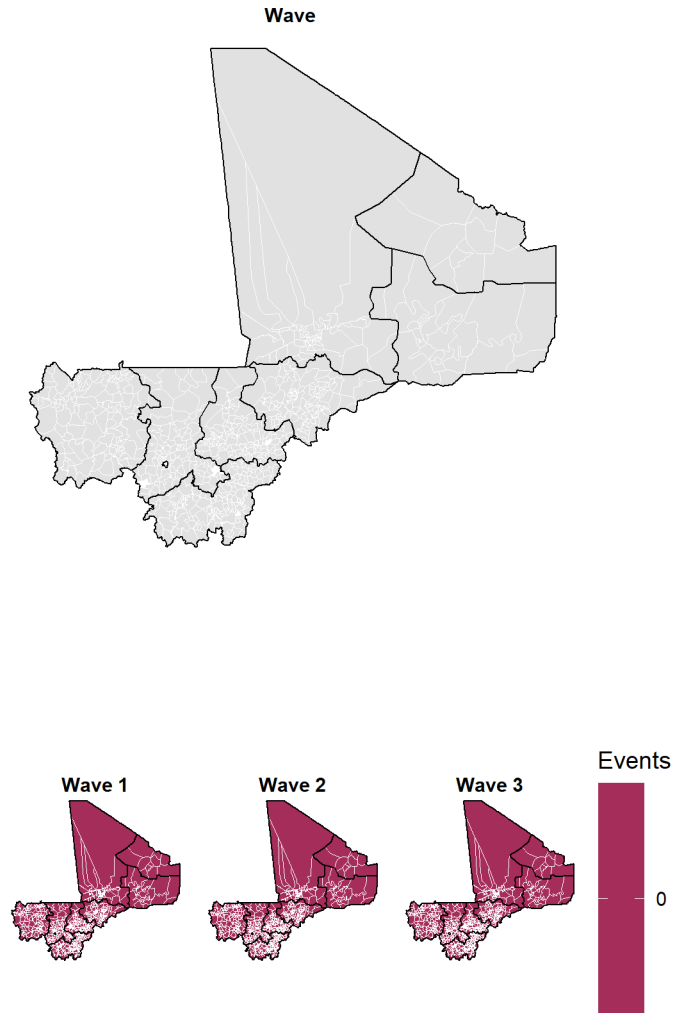


Figure 8: Commune-level (admin-3) favorability from the main model (top, sampled communes only) and ACLED violent events (bottom, all communes) by wave. Region boundaries in black.

Commune names are matched to the boundary file within region, after removing accents, case, and punctuation. Unmatched communes are missing from Figure 8. Resolve them with a lookup table before interpreting the map.

Source	n	examples
Survey communes not on map	276	kayes kayescommune01; kayes kayescommune03; kayes kayescommune04; kayes kayescommune05; kayes kayescommune07
ACLED communes not on map	212	kayes kayescommune01; kayes kayescommune02; kayes kayescommune05; kayes kayescommune08; kayes kayescommune09
Duplicate commune names on map (same region)	4	koulikouro benkadi; koulikouro diedougou; sikasso kapala; segou somo

4 Diagnostics

Table 3: Convergence status for every fitted model. A singular fit means a variance component was estimated at or near zero.

Model	converged	singular	message
M0	TRUE	FALSE	
M1	TRUE	FALSE	
M2	TRUE	FALSE	
M3	TRUE	FALSE	
M4	TRUE	FALSE	
M5	TRUE	FALSE	
M6	TRUE	FALSE	
Log weight	TRUE	FALSE	
Any event	TRUE	FALSE	
Wave 1	TRUE	TRUE	
Wave 2	TRUE	TRUE	
Wave 3	TRUE	FALSE	
Drop Bamako	TRUE	FALSE	
Drop Gao	TRUE	FALSE	
Drop Kayes	TRUE	FALSE	
Drop Koulikoro	TRUE	FALSE	
Drop Mopti	TRUE	FALSE	
Drop Segou	TRUE	FALSE	
Drop Sikasso	TRUE	FALSE	
Drop Tombouctou	TRUE	FALSE	

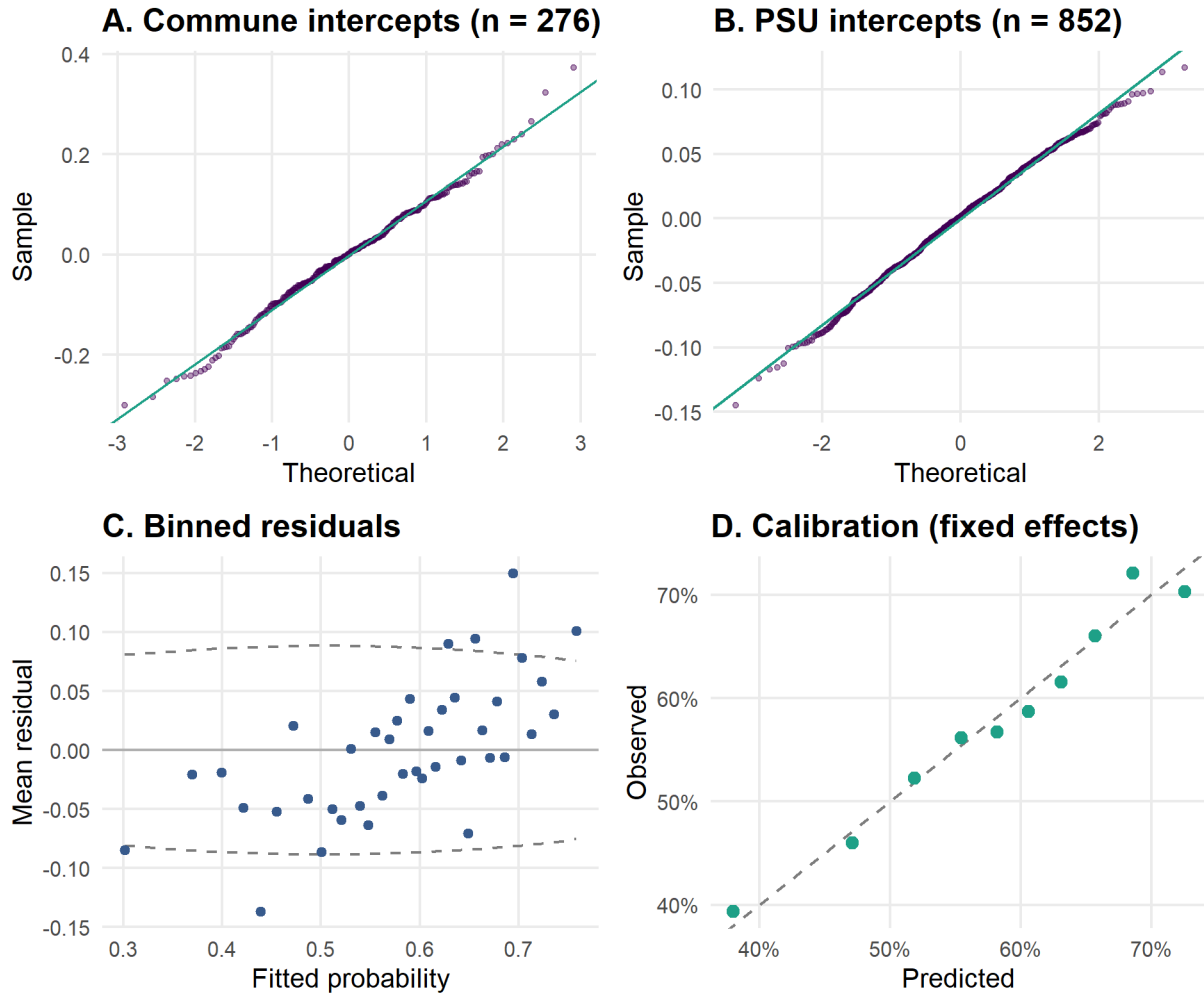


Figure 9: Main-model diagnostics. (A, B) Normal Q-Q plots of estimated random intercepts. (C) Binned residuals: average residual within 40 bins of fitted probability, with approximate 95% bounds. (D) Calibration: observed versus predicted favorability by decile of predicted probability.

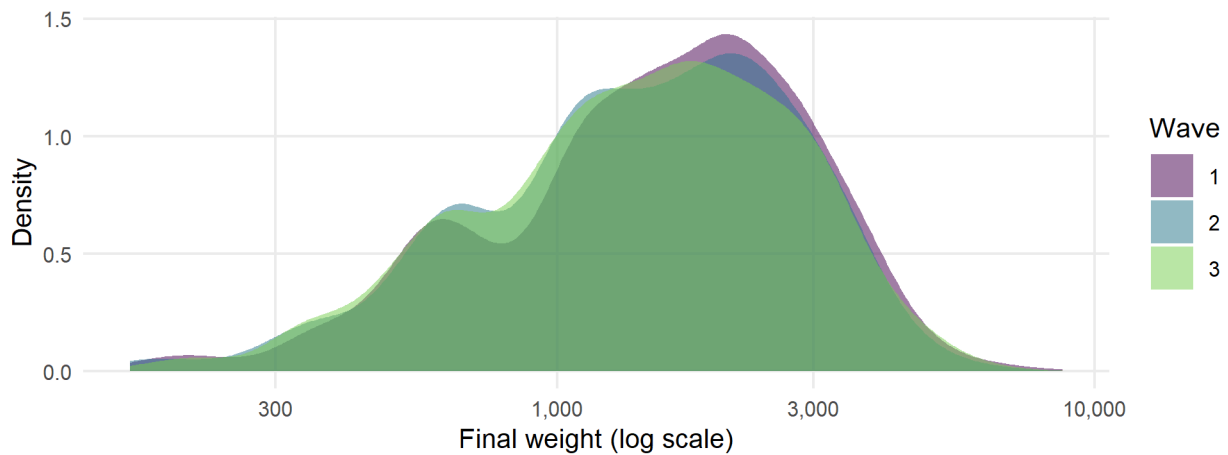


Figure 10: Distribution of final weights by wave (log scale). Long right tails inflate the Kish design effect and the design-based standard errors.

The commune and PSU intercepts should lie close to their reference lines; with communes observed only a few times and roughly 6 respondents per PSU, individual intercepts are heavily shrunk toward zero, so mild tail departures are expected and are not evidence against the model. Binned residuals falling outside the dashed bounds in more than about 5% of bins, or systematic curvature, would point to a missing non-linearity, most often in age or in violence itself.

5 Bayesian fallback

The lme4 main model converged without a singular fit, so the Bayesian fallback was not run (set `cfg$use_brms = TRUE` to run it anyway). It fits M3 with brms using weakly informative priors: Normal(0, 1.5) on coefficients and Student-t(3, 0, 1) on random-effect SDs, which regularize the variance components when many communes contribute few respondents. With 32 cores it runs 4 chains in parallel and, with the cmdstanr backend, threads within each chain.

6 Discussion

6.1 Limitations

The estimates should be read against the main sources of total survey error.

- **Measurement and linkage of the exposure.** ACLED relies on media and partner reporting, which undercounts events in remote and access-restricted areas. If undercounting is concentrated in the most affected northern and central communes, the exposure is measured with error that most likely attenuates the association. Event locations geocoded only to a region or cercle cannot be assigned to a commune, and commune name matching between ACLED and the survey frame can drop events silently (22 ACLED commune codes did not match a sampled commune).
- **Timing.** The exposure window must end before each commune's fieldwork dates. Events during fieldwork can reflect the same conditions that shape responses.
- **Coverage and nonresponse related to violence.** Insecurity affects which PSUs can be fielded, whether replacement PSUs were used, and who is at home. If the most affected places are underrepresented, the estimate describes the accessible population.
- **Measurement of the outcome.** Expressing a view of the leader can be sensitive, and reluctance may itself vary with local insecurity, which would bias the violence association in either direction.
- **Confounding.** The within-commune estimate controls stable commune characteristics, but not time-varying ones such as displacement, military presence, or local economic shocks. The between-commune estimate is more exposed.
- **Mediators.** Attitudes measured alongside the outcome cannot establish mediation.
- **Model-based versus design-based inference.** The main estimates are model-based. They generalize to the sampled population only to the extent that the design is ignorable given the model; the design-based check in Figure 6 is the evidence on that point.

7 References

- Bell, A., & Jones, K. (2015). Explaining fixed effects: Random effects modeling of time-series cross-sectional and panel data. *Political Science Research and Methods*, 3(1), 133-153.
- DuMouchel, W. H., & Duncan, G. J. (1983). Using sample survey weights in multiple regression analyses of stratified samples. *Journal of the American Statistical Association*, 78(383), 535-543.
- Lumley, T. (2010). *Complex surveys: A guide to analysis using R*. Wiley.

Raleigh, C., Linke, A., Hegre, H., & Karlsen, J. (2010). Introducing ACLED: An armed conflict location and event dataset. *Journal of Peace Research*, 47(5), 651-660.

Appendix A: Model formulas

M0: $\text{fav} \sim 1 + (1 \mid \text{commune_id}) + (1 \mid \text{psu_w})$

M1: $\text{fav} \sim 1 + \text{wave} + \text{region} + (1 \mid \text{commune_id}) + (1 \mid \text{psu_w})$

M2: $\text{fav} \sim 1 + \text{wave} + \text{region} + \text{violence_log} + (1 \mid \text{commune_id}) + (1 \mid \text{psu_w})$

M3: $\text{fav} \sim 1 + \text{wave} + \text{region} + \text{violence_log} + \text{female} + \text{age10} + \text{ethnicity} + \text{education} + \text{urban} + (1 \mid \text{commune_id}) + (1 \mid \text{psu_w})$

M4: $\text{fav} \sim 1 + \text{wave} + \text{region} + \text{violence_log} + \text{female} + \text{age10} + \text{ethnicity} + \text{education} + \text{urban} + \text{local_gov_approve} + \text{democracy_c} + (1 \mid \text{commune_id}) + (1 \mid \text{psu_w})$

M5: $\text{fav} \sim 1 + \text{wave} + \text{region} + \text{violence_log} + \text{female} + \text{age10} + \text{ethnicity} + \text{education} + \text{urban} + \text{local_gov_approve} + \text{democracy_c} + \text{perceived_c} + \text{stress_c} + (1 \mid \text{commune_id}) + (1 \mid \text{psu_w})$

M6: $\text{fav} \sim 1 + \text{wave} + \text{region} + \text{female} + \text{age10} + \text{ethnicity} + \text{education} + \text{urban} + \text{violence_between} + \text{violence_within} + (1 \mid \text{commune_id}) + (1 \mid \text{psu_w})$

Appendix B: Main model fixed effects

Term	OR	95% low	95% high	p
(Intercept)	2.24	1.57	3.18	< .001
wave2	1.27	1.10	1.47	0.001
wave3	1.42	1.23	1.65	< .001
regionGao	0.59	0.40	0.88	0.009
regionKayes	0.81	0.57	1.16	0.256
regionKoulikoro	0.66	0.47	0.94	0.020
regionMopti	0.59	0.40	0.86	0.006
regionSegou	0.67	0.47	0.95	0.024
regionSikasso	0.88	0.62	1.26	0.499
regionTombouctou	0.56	0.39	0.81	0.002
violence_log	0.72	0.66	0.79	< .001
female	1.02	0.91	1.15	0.714
age10	1.05	1.00	1.10	0.038
ethnicityFulani	0.78	0.66	0.93	0.006
ethnicitySoninke	1.08	0.88	1.33	0.434
ethnicitySonghai	1.09	0.87	1.38	0.456
ethnicityOther	1.18	0.96	1.46	0.121
ethnicityDogon	1.07	0.85	1.34	0.562
ethnicityTuareg	0.60	0.47	0.78	< .001
educationPrimary	0.96	0.84	1.10	0.573
educationSecondary+	0.83	0.72	0.96	0.011
urban	0.77	0.63	0.93	0.006